EE4-45
EE9-SO22

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE/EIE PART IV: MEng and ACGI

Corrected copy

**WAVELETS AND APPLICATIONS**

Tuesday, 14 May 10:00 am

Time allowed: 3:00 hours

**There are FOUR questions on this paper.**

**Answer ALL questions.**

*All questions carry equal marks.*

**Any special instructions for invigilators and information for candidates are on page 1.**

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Second Marker(s) : A. Manikas

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**Special Information for the Invigilators: NONE**

**Information for Candidates:**

*Sub-sampling by an integer N*

$$x_{\downarrow N}[n] \longleftrightarrow \frac{1}{N}\sum_{k=0}^{N-1} X(e^{j(\omega-2\pi k)/N}) = \frac{1}{N}\sum_{k=0}^{N-1} X(W_N^k z^{1/N}),$$

where

$$W_N^k = e^{-j2\pi k/N}.$$

*Dual Basis:*
Given a basis $\{\varphi_i(t)\}_{i\in\mathbb{Z}}$, the dual basis is given by the set of elements $\{\tilde{\varphi}_i(t)\}_{i\in\mathbb{Z}}$ satisfying:

$$\langle \varphi_i(t), \tilde{\varphi}_j(t)\rangle = \delta_{i,j}.$$

*Frequency shift property*

$$e^{-j\omega_0 t}x(t) \Leftrightarrow \hat{x}(\omega+\omega_0)$$

*Poisson Summation Formula*

$$\sum_{n=-\infty}^{\infty} \varphi(t-nT) = \frac{1}{T}\sum_{k=-\infty}^{\infty} \hat{\varphi}\left(\frac{2\pi k}{T}\right) e^{-j2\pi kt/T}$$

## The Questions

1. (a) Given the filter $g[n]$, you want

$$\langle g[n], g[n-3k] \rangle = \delta_k. \tag{1}$$

By operating in the $z$-domain, derive the conditions the filter $G(z)$ has satisfy for Eq. (1) to hold. [6]

(b) Consider the transmultiplexer shown in Fig. 1a. Three signals are multiplexed and sent over a high bandwidth channel. A perfect reconstruction (PR) multiplexer cancels crosstalk and reconstruct the signals exactly

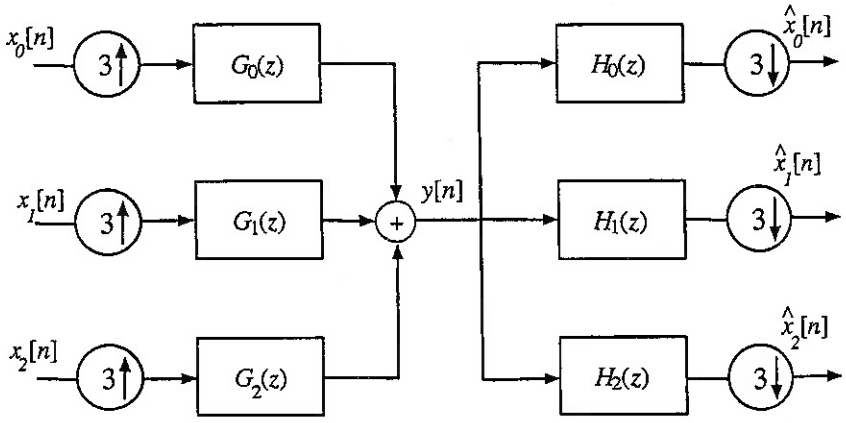



Figure 1a: A transmultiplexer.

i. Give the input/output relations in the z-transform domain. What are the conditions on the filters that guarantee that the transmultiplexer is PR? [8]

ii. Assume that $H_i(z) = G_i(z^{-1})$ for $i = 0, 1, 2$. By using the PR conditions you have just derived and the results in part (a), write, in the time domain, the conditions that $g_0[n], g_1[n]$ and $g_2[n]$ have to satisfy for the transmultiplexer to be PR. [3]

iii. Assume that $H_i(z) = G_i(z^{-1})$ for $i = 0, 1, 2$ and that

$$G_0(z) = (1 + z^{-1} + z^{-2})/\sqrt{3},$$

find the shortest possible $g_1[n]$ and $g_2[n]$ that would satisfy the PR conditions. You also want $g_1[n]$ to be symmetric and $g_2[n]$ to be anti-symmetric. [8]

2. Consider the tree-structured filter bank shown in Fig. 2a.

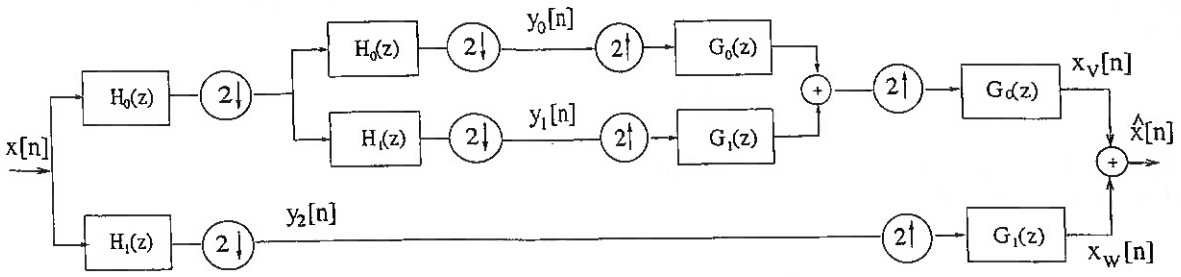



Figure 2a: Tree-structured filter bank.

(a) Let $G_1(z) = \frac{\sqrt{2}}{4(\alpha-1)}\left(z^{-3} + (\alpha-2)z^{-2} - (2\alpha-1)z^{-1} + \alpha\right)$ with $\alpha = 2 - \sqrt{3}$. Design $G_0(z), H_0(z), H_1(z)$ in order to obtain an orthogonal perfect reconstruction filter bank.

[8]

(b) Find the zeros of $G_1(z)$. [Hint: if you correctly guess one of the zeros, you will be left with an easy factorization]

[7]

(c) Assume $x[n] = n + 3$ and ignore any boundary effect. Which of the signals $y_0[n], y_1[n], y_2[n]$ is nonzero? (Justify your answer).

[5]

(d) Assume now that $x[n] = n^2 + 1$ and again ignore any boundary effect. Which of the signals $y_0[n], y_1[n], y_2[n]$ is nonzero? (Justify your answer).

[5]

3. Consider a set of linearly independent functions $\{\varphi_i\}_{i=1,2,..,N}$ that covers the sub-space $V$, that is, $V = span\{\varphi_i\}_{i=1,2,...N}$. Consider a function $f(t) \in L_2(\mathbb{R})$, the orthogonal projection of $f(t)$ onto $V$ is:

$$\hat{f}(t) = \sum_{i=1}^{N} c_i \varphi_i(t),$$

with $c_i = \langle f(t), \tilde{\varphi}_i(t) \rangle$, $i = 1, 2, ...N$. Here $\{\tilde{\varphi}_i\}_{i=1,2,..,N}$ is the dual basis of $\{\varphi_i\}_{i=1,2,..,N}$. Based on the above formula and assuming

$$f(t) = \begin{cases} 2t & \text{for } 0 \leq t < 1/2 \\ 1 & \text{for } 1/2 \leq t < 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) compute the coefficients $c_i$ of the orthogonal projection of $f(t)$ onto the space spanned by $\varphi(t), \psi(t), \sqrt{2}\psi(2t), \sqrt{2}\psi(2t-1)$ with

$$\varphi(t) = \begin{cases} 1 & \text{for } 0 \leq t < 1 \\ 0 & \text{otherwise} \end{cases}$$

and

$$\psi(t) = \begin{cases} 1 & \text{for } 0 \leq t < 1/2 \\ -1 & \text{for } 1/2 \leq t < 1 \\ 0 & \text{otherwise,} \end{cases}$$

[7]

(b) sketch and dimension the function $\hat{f}(t)$,

[6]

(c) compute the energy of the error function $\epsilon(t) = f(t) - \hat{f}(t)$,

[6]

(d) verify that $\|f(t)\|^2 = \|\hat{f}(t)\|^2 + \|\epsilon(t)\|^2$, where $\|x(t)\|^2 = \int_{-\infty}^{\infty} x^2(t)dt$.

[6]

4. Consider an arbitrary real-valued function $\varphi(t)$ with Fourier transform $\hat{\varphi}(\omega)$.

(a) Using the Poisson summation formula, show that a function $\varphi(t)$ satisfies partition of unity:

$$\sum_{k=-\infty}^{\infty} \varphi(t-k) = 1$$

when $\hat{\varphi}(2\pi n) = 1$ for $n = 0$ and $\hat{\varphi}(2\pi n) = 0$ for $n \in \mathbb{Z}$ and $n \neq 0$. [6]

(b) Assume now that $\hat{\varphi}(2\pi n) = 1$ for $n = 0$ and that $\hat{\varphi}^{(i)}(2\pi n) = 0$ for $n \in \mathbb{Z}$ and $n \neq 0$, where $\hat{\varphi}^{(i)}(\omega)$ indicates the $i$-th derivative of $\hat{\varphi}(\omega)$ and $i = 0, 1$. Show that, under the above assumptions, there exists coefficients $c_k$ such that:

$$\sum_{k=-\infty}^{\infty} c_k \varphi(t-k) = t.$$

Hint: use the Poisson summation formula and the fact that

$$\hat{\varphi}^{(1)}(\omega) = -j \int_{-\infty}^{\infty} t\varphi(t) e^{-j\omega t} dt.$$

[9]

(c) Rather than reproducing polynomials, you want $\varphi(t)$ to be able to reproduce exponentials. Show that, there exists coefficients $c_k$ such that:

$$\sum_{k=-\infty}^{\infty} c_k \varphi(t-k) = e^{j\omega_0 t}$$

when $\hat{\varphi}(\omega_0) \neq 0$ and $\hat{\varphi}(\omega_0 + 2\pi n) = 0$ for $n \in \mathbb{Z}$ and $n \neq 0$.
[Hint: assume $c_k = c_0 e^{j\omega_0 k}$ for a properly chosen constant $c_0 \neq 0$.] [10]